



AUV pose correction via underwater object recognition using synthetic data

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Abstract

Accurate localization of autonomous underwater vehicles (AUVs) is challenging because inertial measurement units (IMUs) and Doppler velocity logs (DVLs) accumulate drift during long-duration missions. To address this challenge, this study explores whether synthetic sonar imagery can provide transferable landmark cues for real-world mapping, thereby reducing dependence on costly and limited field datasets. Using an Unreal Engine 5-based FURo-sim framework, we generated forward-looking sonar (FLS) data and automatically annotated high-intensity returns and their acoustic shadows. These annotations are used to train a YOLO11 detector exclusively on synthetic data, and the centroids of the resulting bounding boxes are interpreted as landmark observations. The landmark detections are then fused with DVL/IMU dead reckoning within a two-dimensional pose-graph optimization framework to refine both the vehicle trajectory and the resulting sonar mosaics. The proposed approach is validated using real coastal-sea data, with ultra-short baseline (USBL) acoustic positioning serving as a proxy for ground truth. Experimental results show that the method reduces dead-reckoning localization error by 15.4%, demonstrating the effectiveness of the synthetic data-driven detector-based pose-graph formulation for real-world AUV navigation.

Keywords Sim-to-real · Underwater localization · YOLO11 · Trajectory correction · Pose-graph optimization

1 Introduction

Localization is a fundamental requirement for autonomous underwater vehicle (AUV) missions, as it provides spatial context for sensor data, supports safe navigation, and underpins tasks such as path following and waypoint tracking. In underwater environments, however, global positioning system (GPS) is unavailable, so most operational AUVs rely on dead reckoning based on Doppler velocity log (DVL) and inertial measurement unit (IMU) measurements, often supplemented by external acoustic aids. Although this navigation scheme is self-contained and straightforward to deploy, drift errors inevitably accumulate over time during long-duration missions in the ocean.

To mitigate this drift, exteroceptive measurements from sonar, cameras, and acoustic beacons have been fused with inertial navigation in graph-based localization frameworks for AUVs [1], where stable artificial targets are utilized as landmarks [2].

In many of these systems, pre-existing or deliberately deployed man-made structures such as pipelines and artificial targets are treated as stable landmarks for localization [3, 4]. Landmark-based approaches perform reliably in environ-

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ments where artificial structures are dense and well mapped. However, in natural coastal environments, artificial structures are sparse, only partially known [5], and spatially confined. Consequently, the vehicle can only maintain effective exploration and pose correction within a narrow operational region, fundamentally restricting the extent of the accessible survey area.

Recent advances in deep learning-based approaches have enabled robust object detection and semantic segmentation on sonar imagery, facilitating the use of detected underwater objects as generic landmarks for navigation and inspection [6]. However, the application of deep learning to sonar data remains severely constrained by the limited availability of annotated datasets. Large-scale, publicly available benchmark datasets remain scarce and are costly to curate [7]. These factors create a data bottleneck that hinders the development and deployment of learning-based sonar perception in real-world underwater missions.

To alleviate this data scarcity, a growing body of work has investigated synthetic sonar imagery and sim-to-real transfer. In particular, sonar simulators based on ray-tracing model have been employed to generate acoustic imaging for training object detectors and segmentation networks [8, 9], while adversarial and style-transfer techniques have been applied to bridge the domain gap between synthetic and real sonar imagery [10, 11]. In parallel, semantic SLAM and pose-graph optimization (PGO) frameworks have shown that object-level or bounding-box-level features can be integrated with odometry estimation to improve localization and mapping by modeling vehicle poses as graph nodes and relative motion and landmark observations as edges in a nonlinear least-squares formulation [12–14]. From a navigation taxonomy perspective, such approaches can be categorized as sparse odometry, in which only salient features are extracted and tracked rather than reconstructing a full dense map. Dong et al. [15] proposed an adaptive sonar-inertial framework combining feature matching with inertial navigation system (INS) to suppress drift, while Xu et al. [16] demonstrated stable localization through keyframe-based sonar odometry. The proposed method builds upon this sparse odometry paradigm but extends it by incorporating object-level recognition results—such as seaweed and artificial structures—directly as navigation constraints, providing reliable pose correction even in feature-sparse underwater environments.

Motivated by these developments, we propose an AUV pose correction method based on an object detector trained with synthetic data. To this end, we employ FURo-sim [17], a simulator built on Unreal Engine (UE) 5, to generate synthetic forward-looking sonar (FLS) images together with automatically produced bounding-box annotations suitable for training one-stage object detectors. Subsequently, we train a YOLO11 detector, building on the YOLO series of one-stage real-time object detectors [18], exclusively

on these synthetic fan-shaped FLS images. These synthetic training datasets include both man-made and natural seaweed-like objects. Once trained, the detector is applied to real FLS data collected from sea trials using an AUV. The detected bounding-box centroids are then interpreted as landmark observations and incorporated into a PGO framework to mitigate dead-reckoning drift errors.

The key contributions of this research are summarized as follows:

1. We employ a FURo-sim-based simulation and automatic annotation pipeline to generate realistic FLS imagery and large-scale YOLO format training datasets without any manual labeling effort.
2. We demonstrate that a YOLO11-based sonar object detector trained solely on synthetic data can be successfully applied to real coastal FLS imagery.
3. We formulate a PGO framework that fuses DVL/IMU odometry with landmark constraints derived from detected objects, and we evaluate the resulting localization improvement on coastal field datasets using USBL measurements as a proxy for ground truth.

2 Problem statement

Existing sensor-fusion pipelines for AUV navigation rely primarily on DVL/IMU-based odometry. Although they provide high-rate measurements, accumulated drift over long-term missions and sensitivity to environmental disturbances remain unresolved challenges [19, 20]. Deep learning-based object recognition has recently emerged as a promising alternative for supplying exteroceptive cues for pose correction [21, 22]. However, its application to underwater environments is severely limited by the lack of large-scale, annotated datasets, which are costly and labor-intensive to obtain due to weather constraints, ocean currents, equipment failures, and the need for manual annotation of sonar imagery [23].

This work addresses this data bottleneck by proposing a sim-to-real object-based pose correction framework, in which an object detector is trained solely on synthetic FLS datasets generated in a UE 5-based underwater simulator and deployed on real sonar measurements [24, 25]. The simulator automatically varies AUV pose, FLS configuration, and environmental conditions to generate large-scale data, while annotations are produced without human supervision. Detected bounding boxes from real sonar frames are then integrated as edges in a pose graph to mitigate drift in conventional DVL/IMU odometry. The proposed approach aims to demonstrate that synthetic data alone can provide actionable object-level constraints for real-world underwater navigation. By enabling learning-based pose correction without reliance on extensive real datasets, this work expands

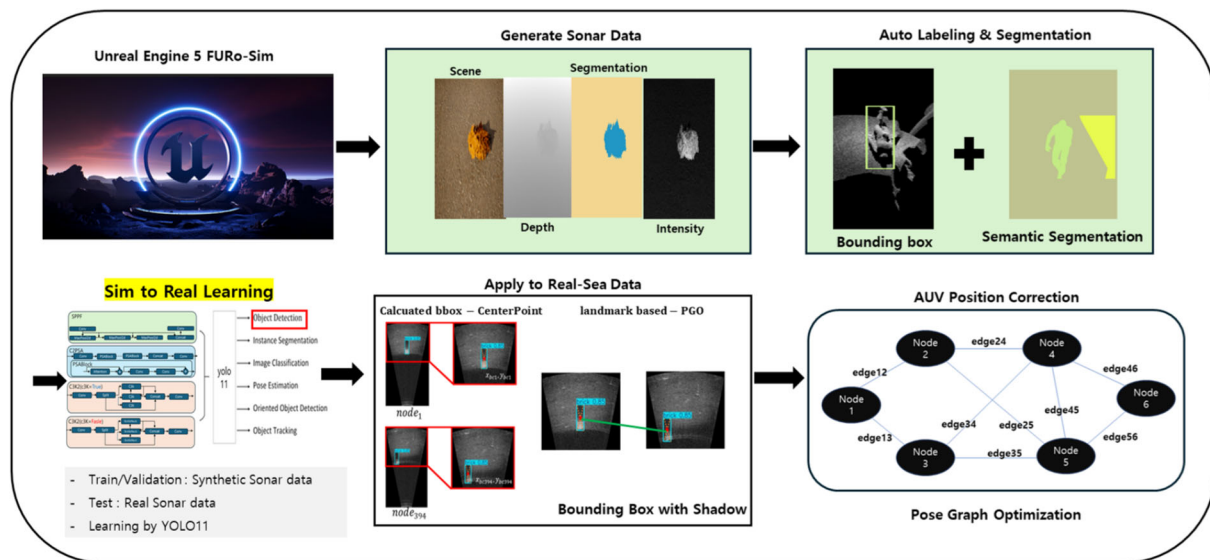


Fig. 1 Proposed pipeline for synthetic sonar image-based AUV localization correction

the applicability of deep learning navigation methods in data-scarce underwater scenarios and provides a scalable direction for robust AUV navigation.

3 Method

The proposed method (Fig. 1) begins with the generation of synthetic sonar images in UE 5-based simulator, along with the creation of bounding-box annotations that include both high-intensity returns and their corresponding acoustic shadows. These synthetic annotated images serve as the training data for a YOLO11 detector. The training and validation sets consist exclusively of synthetic FLS images, whereas real underwater FLS data are reserved solely for testing. During deployment, the trained detector is applied to real FLS data to obtain bounding boxes, from which landmark centroids are extracted. The relative displacements of centroids corresponding to the same landmark across consecutive frames are then incorporated as additional constraints in the pose graph. Overall, by augmenting DVL/IMU dead reckoning with landmark constraints derived from a simulation-trained YOLO11 detector, the proposed framework aims to improve mosaic alignment and trajectory consistency without requiring any additional external navigation sensors.

3.1 Synthetic data generation

To support the proposed framework, synthetic FLS data were generated using FURo-sim, a simulation environment built on UE 5. Notably, this platform enables the construction of an automatic annotation pipeline for training the YOLO11 detector, aligning with recent studies that adopt

game-engine-based sonar simulators for dataset creation [26]. As illustrated in Fig. 2, each rendered camera image in the simulator is first converted into a set of intermediate representations, namely depth, intensity, and semantic segmentation maps. These outputs are subsequently combined to synthesize a sonar-style intensity image along with its associated bounding-box and segmentation labels.

In the virtual scene, an FLS is rigidly mounted on the vehicle, and at each rendering step the simulator produces a depth image $D(u, v)$, a surface-normal map $\mathbf{n}(u, v)$, a semantic segmentation map $s(u, v)$, and an acoustic intensity image $I(u, v)$ on an image grid of width W and height H . The slant range at each pixel is directly obtained from the corresponding value in the rendered depth image $D(u, v)$:

$$r(u, v) = D(u, v). \quad (1)$$

The vertical and horizontal beam angles associated with pixel (u, v) are parameterized using the vertical and horizontal fields of view, Θ_v and Θ_h , as

$$\theta_v(v) = \Theta_v \left(\frac{1}{2} - \frac{v}{H} \right), \quad (2)$$

$$\theta_h(u) = \Theta_h \left(\frac{1}{2} - \frac{u}{W} \right). \quad (3)$$

The unit ray direction $\ell(u, v)$ in the sonar frame is obtained by rotating the forward axis according to $(\theta_h(u), \theta_v(v))$ and applying the fixed extrinsic transform between the vehicle and the sensor. Given $\ell(u, v)$ and $r(u, v)$, the simulator internally computes the acoustic intensity $I(u, v)$ using a GPU-based sonar rendering model.

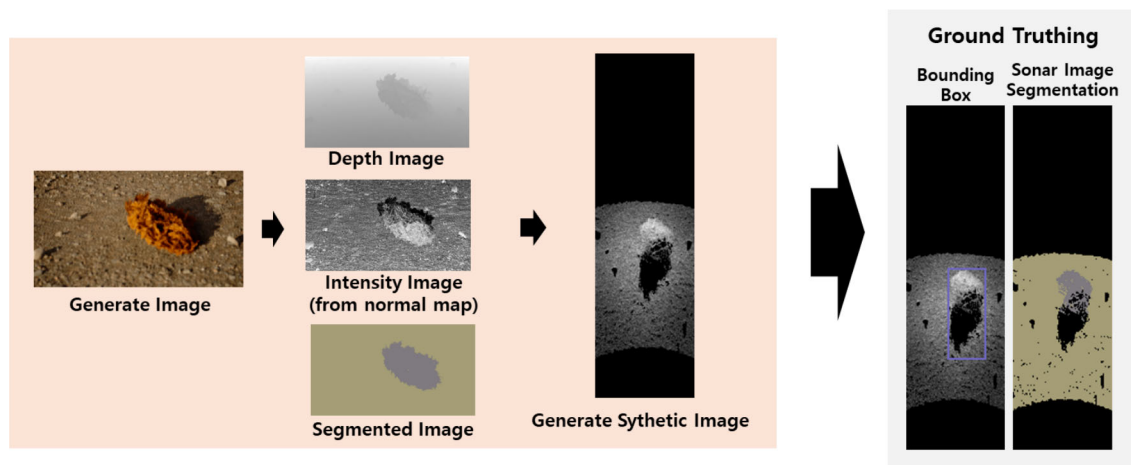


Fig. 2 Synthetic FLS image generation using FURo-sim

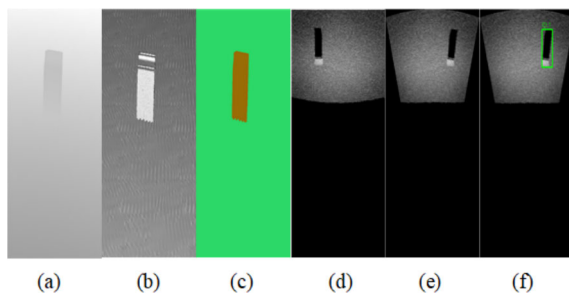


Fig. 3 Data generation pipeline from simulator outputs to YOLO labels. Panels (a)–(f) show, respectively, the depth image, intensity image, semantic segmentation mask, polar sonar image on the range-beam grid, fan-shaped sonar projection, and bounding-box annotation

The resulting continuous angles and ranges are then discretized onto a 512×96 range-beam lattice by linearly mapping (θ_h, r) into integer beam and range indices, and all intensity samples $I(u, v)$ that fall into the same cell are accumulated. This raw range-beam image is subsequently remapped into a polar fan-shaped image of size 300×456 , which serves as the canonical sonar image plane in our method.

In this representation, objects are arranged on a regular two-dimensional grid compatible with standard YOLO-style bounding boxes, and both synthetic and real FLS frames are converted into the 300×456 fan format so that the detector can be trained and evaluated within a common image domain. A representative example of this conversion pipeline—from depth and intensity images to semantic segmentation, polar and fan-shaped sonar projections, and the final YOLO bounding-box labels—is shown in Fig. 3.

In addition to depth and intensity images, the simulator also generates a pixel-wise semantic segmentation map that is spatially aligned with the rendered images. Each pixel encodes an integer class label, which is subsequently con-

verted into a semantic mask on the camera image plane. To incorporate these masks into the synthetic sonar domain, each pixel is mapped to a range-beam cell based on its depth and beam direction, ensuring that the segmentation is precisely aligned with both the 512×96 polar grid and the derived fan-shaped image. Although the native range-beam lattice could, in principle, directly serve as an input to the detector, we adopt the fan-shaped representation as the canonical sonar image plane. This choice provides a more intuitive structure for both human interpretation and learning-based detection, while aligning range and cross-range along orthogonal axes. In this domain, the width and height of YOLO-style axis-aligned bounding boxes directly correspond to the target's cross-range extent and the along-range length of its highlight-shadow pattern. Consequently, the resulting centroid displacements can be consistently mapped to metric variations in range and bearing, enabling their use in the PGO framework.

For each target class, we extracted connected components in the segmentation map and computed tight axis-aligned bounding boxes that covered the bright highlight regions. To incorporate the acoustic shadow, each box was extended in the range direction. Specifically, the far-range boundary was first updated using the maximum depth value within the component and then further extended by sampling rows beyond the highlight along the same beam direction. This design mirrors the typical appearance of real FLS targets, which consist of a strong return followed by a trailing acoustic shadow that encodes additional information about shape and range. The resulting near and far ranges were mapped to range indices on the polar grid, yielding rectangular regions that enclosed both the primary highlight and its trailing shadow. Within these regions, empty sonar pixels were filled with the corresponding class label to obtain a shadow-augmented segmentation mask. This mask was then converted to the fan-shaped domain, from which axis-aligned bounding boxes

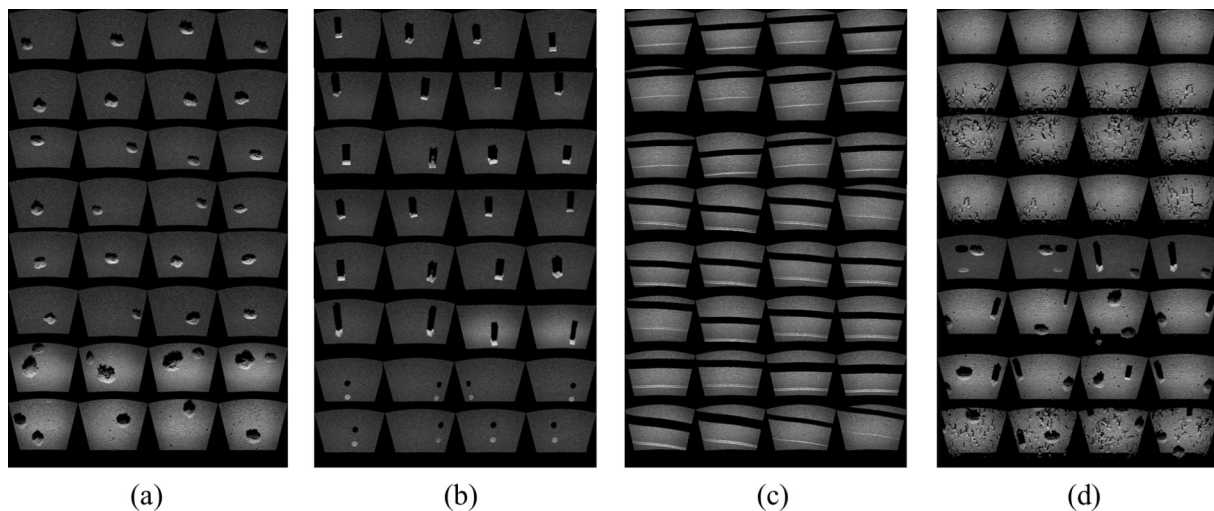


Fig. 4 Examples of synthetic images. Panels: **(a)** natural seaweed objects, **(b)** artificial objects (brick and cylinder), **(c)** the dummy class (e.g., ropes), and **(d)** background-only and mixed-class synthetic images where different object classes appear together

were extracted and stored in the YOLO format. By explicitly including both highlight and shadow in the annotation, the detector was encouraged to learn the composite appearance of sonar targets rather than relying solely on isolated bright spots, which improved robustness when transferring to real FLS imagery. This procedure allowed the simulator to automatically generate training annotations whose bounding boxes consistently captured both target highlights and their associated acoustic shadows.

3.2 Object detection

We trained a YOLO11 detector using automatically generated training annotations derived from synthetic FLS images. The network was optimized for 1000 epochs with an input resolution of 480 and a batch size of 8, while all other hyperparameters followed the default Ultralytics configuration.

Ultralytics YOLO11 employs an anchor-free detection head, in which each location on the output feature maps directly predicts the bounding-box coordinates, an objectness score, and class probabilities without relying on predefined anchor boxes. Each detected object is parameterized by the center coordinates (x, y) of the bounding box on the image plane together with its width and height. For training, all bounding boxes are converted into the normalized YOLO format with respect to the image width W and height H . When rectangular network inputs are formed via resizing or padding, the same transformations are consistently applied to the labels so as to preserve geometric correctness.

Three semantic classes were defined: a seaweed class representing natural landmarks, an artificial object class consisting of bricks and cylinders representing artificial landmarks, and a dummy class corresponding to rope segments.

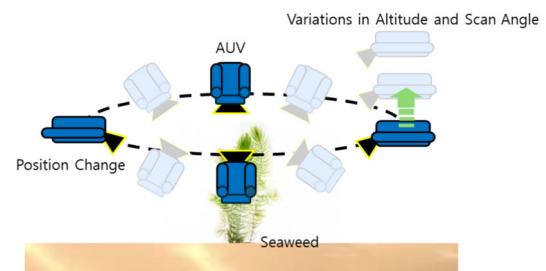
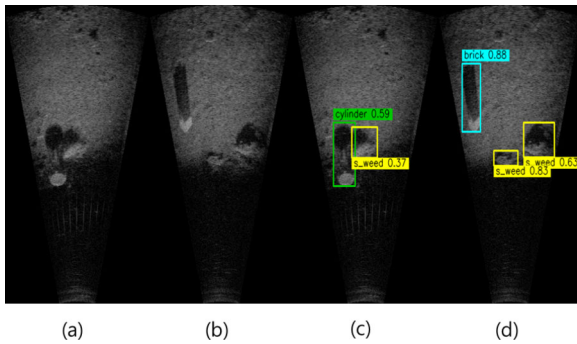


Fig. 5 Synthetic data generation under various AUV poses, sensor look-angles, and altitudes relative to a seaweed object

The dummy class is used to reduce false positives on rope-shaped structures. These three classes were intentionally selected because each exhibits acoustically distinct characteristics in sonar images: bricks produce strong specular reflections from flat surfaces, cylinders generate uniform cylindrical scattering patterns, and seaweed produces irregular multi-path scattering. This inherent acoustic separability between classes ensures minimal cross-class confusion during detection. Figure 4 (a)–(c) shows seaweed-like natural landmarks, bricks and cylinders, and rope-based dummy classes. (d) presents background-only scenes (in the upper four rows) and mixed-class scenes, where different objects appear together within the same scene (in the lower four rows). To ensure diversity across environmental conditions, the dataset was constructed by combining multiple scan angles, five seaweed morphologies, three range levels, and three noise levels. The synthetic sonar dataset was constructed by varying the robot pose and heading, the tilt angle of the FLS, the vehicle altitude, the level of speckle and background noise, and the size and placement of underwater objects (Fig. 5). The final dataset comprised 731 training images (221 seaweeds, 399 artificial objects, 111 dummies),

Table 1 Quantitative detection performance on real FLS data

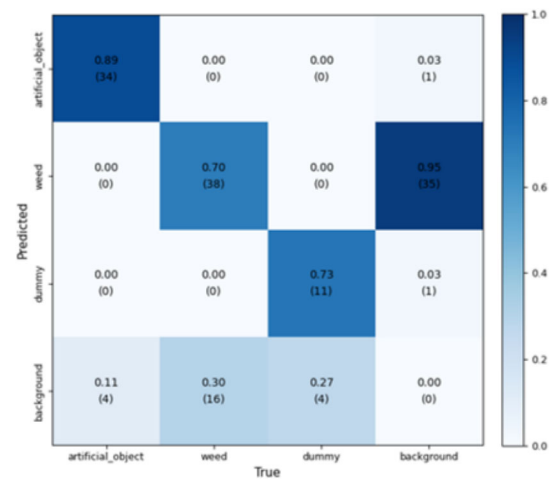
Metric	Value
Precision	0.692
Recall	0.776
F1-score	0.731
mAP@0.5	0.813

**Fig. 6** Representative detection results using real FLS images. (a), (b) input fan-shaped sonar images; (c), (d) YOLO11 detections of brick, cylinder, and seaweed

182 validation images (51 seaweeds, 100 artificial objects, 31 dummies), and 107 test images (54 seaweeds, 38 artificial objects, 15 dummies).

To evaluate the object detection performance on real FLS data quantitatively, we analyzed precision, recall, F1-score, and mean average precision (mAP)@0.5. As presented in Table 1, the proposed detector achieved a precision of 0.692, a recall of 0.776, and an F1-score of 0.731, with an mAP@0.5 of 0.813. Representative detection results on real FLS imagery are shown in (Fig. 6). To further examine the detection characteristics for each semantic category, illustrates the normalized confusion matrix. The horizontal axis represents the actual ground truth classes, while the vertical axis indicates the predicted classes. Each column is normalized to show the proportion of how a specific ground truth class was predicted, with the corresponding raw sample counts indicated in each cell. The diagonal elements correspond to true positives, where the recall rates for the artificial object, seaweed, and dummy classes were 89%, 70%, and 73%, respectively. Off-diagonal elements represent class confusion. For instance, cases where the background was misidentified as seaweed were relatively frequent. Instances where the predicted class is categorized as background indicate objects that were present but not detected, effectively representing false negatives absorbed into the background row (Fig. 7).

To use the detector outputs as landmarks, each detected bounding-box center is interpreted as a two-dimensional

**Fig. 7** Normalized confusion matrix of the trained model for the artificial object, seaweed, and dummy classes, with the corresponding raw sample counts indicated in each cell

landmark measurement $\mathbf{z}_k^n$ as follows

$$\mathbf{z}_k^n = (x_k^n, y_k^n, w_k^n, h_k^n). \quad (4)$$

The normalized centroid (x_k^n, y_k^n) and size (w_k^n, h_k^n) of the n -th detection in frame k are obtained from x/W , y/H , w/W , h/H , where W and H denote the image width and height in pixels, respectively. Data association is subsequently performed based on the location of landmarks in consecutive image frames.

For every pair of adjacent frames $(k-1, k)$, we first rejected low-confidence detections and abnormally large clutters (e.g., overextended seaweed patches), and then grouped the remaining detections as corresponding candidates. We computed the Euclidean distance between the candidates m and n across consecutive frames (Eq. 5) as follows

$$d_{nm} = \|(x_k^n, y_k^n) - (x_{k-1}^m, y_{k-1}^m)\|_2. \quad (5)$$

All candidate pairs (n, m) were then sorted in ascending order and greedily established one-to-one correspondences by iterative acceptance of the closest pair whose two detections had not yet been matched. A candidate pair was accepted only if it satisfied two consistency checks. First, a size-consistency constraint was imposed so that the apparent scale of the object did not change abruptly between frames:

$$\left| \frac{w_k^n}{w_{k-1}^m} - 1 \right| \leq \varepsilon, \quad \left| \frac{h_k^n}{h_{k-1}^m} - 1 \right| \leq \varepsilon, \quad (6)$$

where $\varepsilon > 0$ is a tolerance parameter that controls the maximum allowed relative change in width and height (we set $\varepsilon = 0.2$ in our experiments). Equation (6) was used to

ensure that the bounding-box size remained similar. Second, a spatial-proximity constraint was imposed by requiring that the center distance did not exceed the diagonal length of the bounding box:

$$d_{nm} \leq \gamma \sqrt{(w_k^n)^2 + (h_k^n)^2}, \quad (7)$$

where $\gamma > 0$ is a tolerance parameter that determines how far the object was allowed to move between adjacent frames relative to its size (we set $\gamma = 0.8$ in our implementation). The pairs that satisfied both Eqs. (6) and (7) were regarded as repeated observations of the same landmark across frames.

The pixel displacements between the two detections of class c were computed as

$$\begin{aligned} \Delta u &= (x_k^n - x_{k-1}^m)W, \\ \Delta v &= (y_k^n - y_{k-1}^m)H, \end{aligned} \quad (8)$$

and subsequently converted into metric units using η_x and η_y (meters per pixel) as

$$\Delta x_k^c = \eta_x \cdot \Delta u, \quad \Delta y_k^c = \eta_y \cdot \Delta v. \quad (9)$$

Here, Δx_k^c and Δy_k^c in Eq. (9) represent the relative motion of the landmark, expressed in meters, along the range and cross-range axes, respectively. The parameters η_x and η_y are fixed conversion factors used to transform pixel displacements into metric units. These factors are determined based on the physical specifications of the FLS, including the detection range, field of view (FOV), and the resolution of the rendered image grid. In this study, consistent values of $\eta_x = 0.0127$ m/pixel and $\eta_y = 0.0097$ m/pixel were applied across all experiments to maintain a uniform distance scale. Finally, the resulting planar displacement was interpreted as a relative motion measurement in the vehicle frame as follows

$$\mathbf{u}_k^c = \begin{bmatrix} \Delta x_k^c \\ \Delta y_k^c \end{bmatrix}, \quad (10)$$

The YOLO-based landmark constraint did not provide direct information about the vehicle heading. The heading increment $\Delta\theta_{ij}$ was instead obtained independently from the IMU yaw estimates and incorporated into the odometry constraint, rather than into the detection-derived measurement. Within the pose graph, $\mathbf{u}_k^c$ in Eq. (10) was therefore represented as a purely translational quantity and was used to introduce additional edges between the corresponding poses (nodes).

3.3 Pose-graph framework

We implemented PGO for AUV trajectory optimization over a sequence of robot poses. We assumed that precise depth

information is obtained from the pressure sensor, thus the robot pose at time step k is represented as a two-dimensional state

$$\mathbf{x}_k = \begin{bmatrix} x_k \\ y_k \\ \theta_k \end{bmatrix}, \quad (11)$$

where x_k and y_k denote the position in the horizontal plane, and θ_k denotes the heading angle. Each pose $\mathbf{x}_k$ is treated as an element of the special Euclidean group $SE(2)$, the group of rigid-body transformations in the plane, and is modeled as a node in the pose graph.

In this formulation, nodes correspond to discrete AUV poses along the trajectory, while edges encode relative pose constraints derived from sensor measurements. These constraints incorporate both incremental motion estimates from DVL and IMU dead reckoning and additional observations derived from sonar-based object detections. All constraints are jointly optimized to obtain a globally consistent estimate of the vehicle trajectory.

3.3.1 Odometry constraint

The odometry constraint between consecutive poses was derived from the DVL and IMU measurements. For a pair of successive nodes i and j , the DVL/IMU-based relative pose measurement is defined as

$$\mathbf{z}_{ij}^{\text{odom}} = \begin{bmatrix} \Delta x_{ij} \\ \Delta y_{ij} \\ \Delta \theta_{ij} \end{bmatrix}, \quad (12)$$

where Δx_{ij} and Δy_{ij} denote the relative translation in the horizontal plane, and $\Delta \theta_{ij}$ denotes the relative heading change from pose i to pose j .

Let $\mathbf{x}_i$ and $\mathbf{x}_j$ denote the pose estimates of nodes i and j in $SE(2)$, respectively. The corresponding predicted relative pose calculated from the current estimates is expressed as

$$\hat{\mathbf{z}}_{ij} = \begin{bmatrix} \Delta \hat{x}_{ij} \\ \Delta \hat{y}_{ij} \\ \Delta \hat{\theta}_{ij} \end{bmatrix}. \quad (13)$$

The corresponding residual is defined as

$$\mathbf{e}_{ij}^{\text{odom}} = \mathbf{z}_{ij}^{\text{odom}} - \hat{\mathbf{z}}_{ij}. \quad (14)$$

This residual was incorporated into the global PGO to enforce consistency between adjacent poses on the basis of the DVL and IMU odometry.

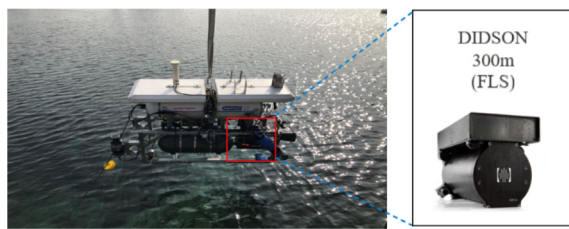


Fig. 8 Sensor configuration on the hovering-type AUV platform

3.3.2 Object constraint

The predicted measurement is calculated as

$$\hat{\mathbf{u}}_{ij} = \begin{bmatrix} \Delta \hat{x}_{ij} \\ \Delta \hat{y}_{ij} \end{bmatrix}. \quad (15)$$

Additional constraints in the pose graph were constructed from the detection-based relative motion measurements $\mathbf{u}_{ij}^c$ defined in Eq. (10). For each object that was consistently detected across consecutive frames, the sequence of centroid positions in the sonar image was tracked and associated over time. The detection residual is then defined as

$$\mathbf{e}_{ij}^{\text{det}} = \mathbf{u}_{ij}^c - \hat{\mathbf{u}}_{ij}, \quad (16)$$

and was used to construct additional edges in the pose graph. In the pose graph, each $\mathbf{u}_{ij}^c$ was incorporated as a purely translational constraint between the corresponding nodes i and j , complementing the odometry constraint $\mathbf{z}_{ij}^{\text{odom}}$. As discussed in Sect. 3.2, the detector did not provide any direct information about the relative heading; the yaw increment $\Delta\theta_{ij}$ was obtained independently from the IMU and was encoded solely in the odometry term.

These detection edges were incorporated together with the full three-degree-of-freedom odometry constraints, providing complementary geometric information that helped to reduce drift and improve global trajectory consistency, while the heading component remained supplied exclusively by the IMU/DVL odometry.

3.3.3 Optimization

The overall PGO problem was formulated as a nonlinear least-squares estimation over all poses $\{\mathbf{x}_{1:k}\}$ and their associated measurement edges. Using the odometry residuals $\mathbf{e}_{ij}^{\text{odom}}$ in Eq. (14) and the detection residuals $\mathbf{e}_{ij}^{\text{det}}$ in Eq. (16), the objective function is written as

$$\min_{\{\mathbf{x}_{1:k}\}} \sum_{(i,j) \in \mathcal{E}_{\text{odom}}} \mathbf{e}_{ij}^{\text{odom}T} \Omega_{ij}^{\text{odom}} \mathbf{e}_{ij}^{\text{odom}} + \sum_{(i,j) \in \mathcal{E}_{\text{det}}} \mathbf{e}_{ij}^{\text{det}T} \Omega_{ij}^{\text{det}} \mathbf{e}_{ij}^{\text{det}}, \quad (17)$$

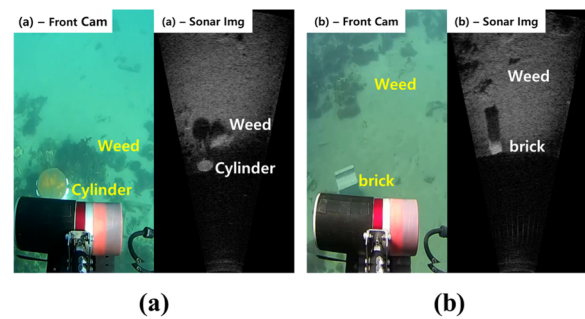


Fig. 9 Examples of optical camera images and their corresponding sonar images acquired during sea trials. In (a), both a cylinder and seaweed are captured, while (b) shows a brick together with seaweed

where $\Omega_{ij}^{\text{odom}}$ and Ω_{ij}^{det} denote the information matrices associated with the odometry and detection constraints, respectively.

The information matrices were defined as the inverse of the covariance matrices. For the odometry constraint, comparable weights were assigned to both translational and rotational components. For the detection-based constraint, higher weights were applied to the translational components, while negligible weights were assigned to the rotational components due to the absence of direct rotational observability. The relative scaling between constraints was determined based on empirical noise characteristics observed in experimental trials.

4 Validation

4.1 Experimental setup

The proposed method was validated with a field test dataset provided by POSTECH. A hovering-type AUV platform equipped with an FLS, a DVL, and an IMU was used for data collection. An overview of the vehicle and mounting configuration of the DIDSON FLS is provided in Fig. 8. Table 2 summarizes the main mechanical characteristics and onboard sensors of the platform, while Table 3 presents the principal specifications of the DIDSON imaging sonar used as the FLS. All sensors were rigidly mounted onto the vehicle frame and operated under a unified time-synchronization scheme.

The AUV traversed approximately $10 \text{ m} \times 8 \text{ m}$ with continuous data acquisition from FLS, DVL, IMU, and USBL. A part of the experimental scene is illustrated in Fig. 9, where each panel presents a front camera image alongside its corresponding FLS sonar image. In (a), both a cylinder and seaweed are captured, while (b) shows a brick together with seaweed.

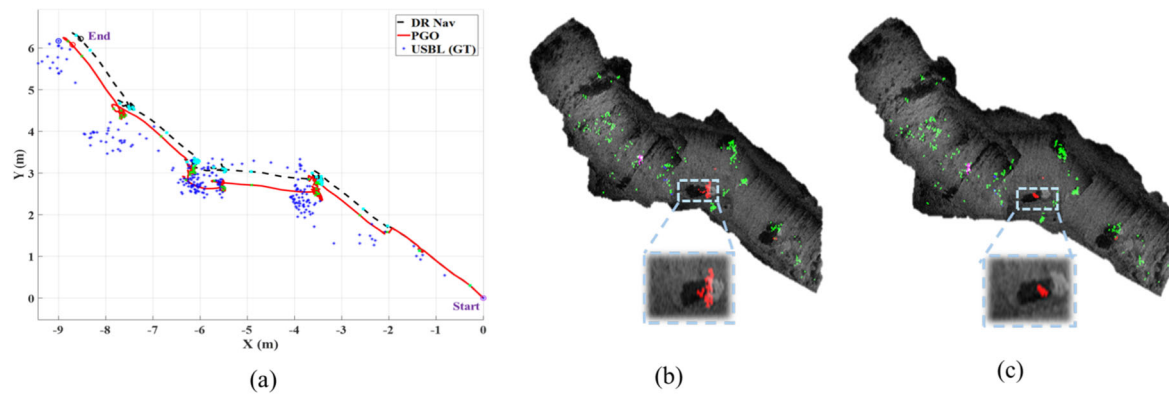


Fig. 10 Validation results of the proposed method. **(a)** Trajectory comparison between DVL/IMU dead-reckoning (dashed black) and PGO-corrected estimate (solid red), with USBL measurements as ground truth (blue dots), **(b)** Mosaicing result before pose correction, **(c)** Mosaicing result after pose correction. In **(a)**, cyan and green markers indicate DR Nav and PGO trajectory nodes at regular intervals, respec-

tively. A purple circle marks the common start point, and black, red, and blue circles denote the end-points of DR Nav, PGO, and USBL, respectively. In **(b)** and **(c)**, green, red, and magenta dots indicate detected seaweeds, artificial objects, and dummies (ropes), respectively. Dashed rectangles highlight regions of significant improvement after pose correction

Table 2 Specifications of the AUV platform used in field experiments

Parameter	Specification
Type	Hovering
Weight (kg)	210 (air)
Dimension (m)	0.9 × 1.5 × 0.9
Depth rating (m)	100
Propulsion (W)	475 (8 thrusters)
Max. speed (m/s)	1.029
Sensors	Doppler Velocity Log Dual frequency imaging sonar Ultra-short baseline Digital pressure transducer Video camera Fiber-optic Gyro

Table 3 Principal specifications of the imaging sonar (DIDSON) used in the present study

Parameter	Specification
Operating frequency	1.8 MHz
Range	0.4–10 m
Number of beams	96
Field of view	29°
Beam spacing	0.3°
Spreading angle	14°
Frame rate	4–21 fps

4.2 Results

The raw USBL measurements were filtered to reject outliers using a Voronoi-diagram-based outlier rejection method [27], and the filtered USBL data were then used as a proxy for ground truth. Figure 10 provides an overview of the validation results. Figure 10 (a) shows the DVL/IMU dead-reckoning trajectory (dashed black) and the PGO-corrected trajectory (solid red), with USBL data presented in blue dots. Figure 10 (b) and (c) present sonar mosaics constructed before and after PGO, respectively. Green, red, and magenta dots indicate the centroids of detected seaweeds, artificial objects, and dummies (ropes), respectively. The dashed rectangles highlight regions where the most significant changes and improvements are observed after pose correction. The red points that were originally widely dispersed within the dashed box converged into a compact cluster after the pose correction.

For a quantitative comparison, we computed the root-mean-square error (RMSE) between each estimated trajectory and the USBL positions. As a result, the dead-reckoning navigation trajectory yielded an RMSE of 0.716 m, and the PGO trajectory achieved an RMSE of 0.604 m with an improvement of approximately 15.4%. Furthermore, to evaluate the long-term drift suppression, we analyzed the end-point error. As summarized in Table 4, the dead-reckoning trajectory resulted in an end-point error of 0.472 m, whereas the PGO-corrected trajectory reduced this to 0.309 m, yielding a 34.5% improvement.

We further compared frame-wise absolute position errors (Fig. 11). Figure 11 (a) shows the absolute x -position error $|e_x|$ with respect to the USBL positions, and (b) presents the corresponding absolute y -position error $|e_y|$. Overall, the

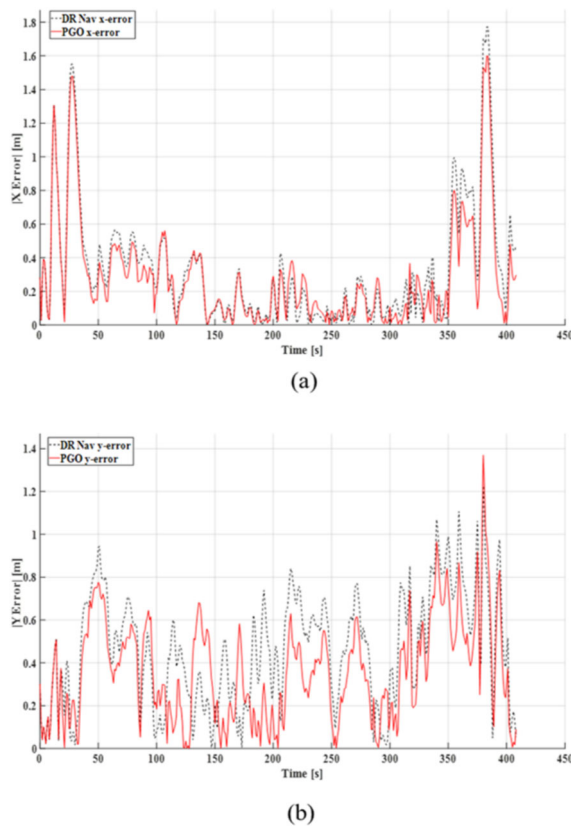


Fig. 11 (a) Absolute x position errors $|e_x|$, and (b) absolute y position errors $|e_y|$ comparing the dead-reckoning trajectory (dashed black) and the optimized estimate (solid red)

Table 4 Quantitative comparison of navigation end-point error

Method	End-point Error (m)
Dead reckoning (DR Nav)	0.472
PGO	0.309
Reduction Rate	34.5%

errors in the y direction are larger than those in the x direction but exhibit a more pronounced decrease after PGO, reflecting the importance of lateral corrections in this coastal survey geometry. Taken together, these results validate the effectiveness of the proposed detector-based pose-graph formulation, demonstrating that it provides reliable and consistent pose corrections on real coastal data despite being trained solely on synthetic sonar images.

4.3 Robustness under DR variations

To further evaluate the robustness of the proposed framework against sensor uncertainty, cumulative Gaussian noise was intentionally introduced into the sway velocity integration to simulate sensor drift. Two additional test cases were generated with increasing noise levels ($\sigma_1 = 1 \times 10^{-4}$ and

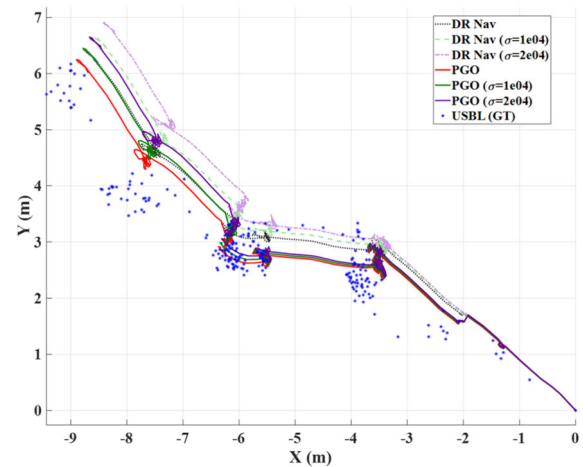


Fig. 12 Trajectory comparison under various DR conditions with USBL measurements as ground truth. The black dotted line indicates DR Nav. The light green and light purple dashed lines represent DR Nav with intentionally introduced cumulative sway velocity errors of $\sigma = 1 \times 10^{-4}$ and $\sigma = 2 \times 10^{-4}$, respectively. The red, green, and purple solid lines represent the corresponding PGO-corrected trajectories for each case. Blue dots indicate USBL ground truth measurements

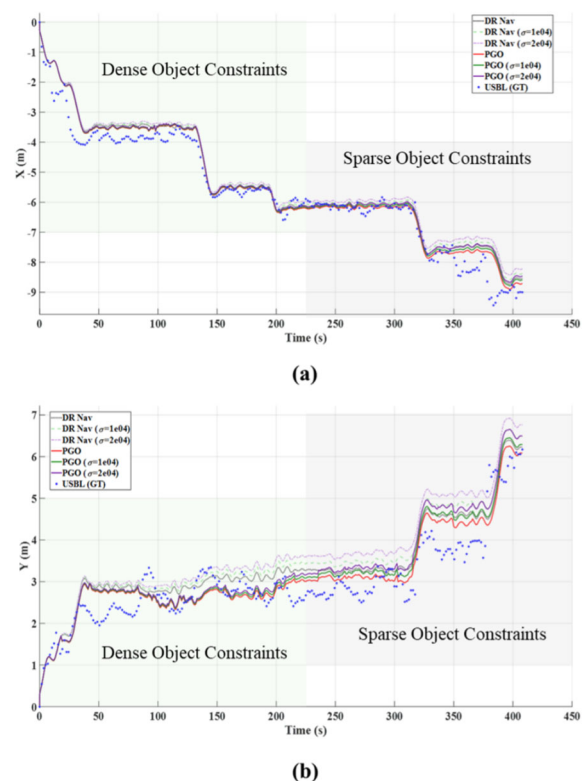


Fig. 13 Comparison of (a) X and (b) Y position estimates over time under various DR conditions. The black dotted line indicates DR Nav. The light green and light purple dashed lines represent DR Nav with intentionally introduced cumulative sway velocity errors of $\sigma = 1 \times 10^{-4}$ and $\sigma = 2 \times 10^{-4}$, respectively. The red, green, and purple solid lines represent the corresponding PGO-corrected trajectories for each case. Blue dots indicate USBL ground truth measurements

$\sigma_2 = 2 \times 10^{-4}$). The proposed PGO framework was applied to all cases, and the results demonstrate consistent performance under all conditions. Quantitatively, for σ_1 , the DR Nav RMSE of 0.841 m is reduced to 0.664 m, corresponding to an improvement of approximately 21.2%. For σ_2 , the RMSE is reduced from 0.987 m to 0.736 m, representing an improvement of approximately 25.4%. In Fig. 12, the black dotted line denotes DR Nav, while the light green and light purple dashed lines represent DR Nav under σ_1 and σ_2 conditions, respectively. The corresponding PGO trajectories are shown as red, green, and purple solid lines. Despite the increasing drift introduced by the Gaussian noise, all PGO trajectories exhibit a consistent corrective trend and converge toward the USBL ground truth, confirming that the proposed framework maintains robust correction performance even under degraded sensor conditions.

A more detailed analysis was conducted by examining the X and Y position estimates over time and comparing them with the USBL ground truth before and after applying PGO. The trajectory is divided into two regions based on the availability of object constraints. The first half corresponds to a dense object constraints region, where sufficient object edges are available to construct reliable constraints, while the latter half represents a sparse object constraints region with significantly limited features. In the X direction, the PGO trajectories remain closely aligned across all conditions throughout the trajectory, with only minor deviations observed toward the end. In the Y direction, the PGO trajectories show strong convergence within the dense constraint region. As the trajectory progresses into the sparse constraint region, gradual divergence appears in the noise-corrupted cases; however, the PGO results effectively suppress this divergence, maintaining bounded deviations even in the absence of sufficient object constraints. These results demonstrate the robustness of the proposed framework in limiting drift accumulation under challenging conditions. The results are presented in Fig. 13, where (a) and (b) illustrate the X and Y position estimates over time, respectively. The black dotted line denotes DR Nav, the light green and light purple dashed lines represent DR Nav under σ_1 and σ_2 conditions, respectively, and the corresponding PGO trajectories are shown as red, green, and purple solid lines, with blue dots indicating USBL ground truth measurements.

5 Discussion

The results presented above indicate that the proposed framework can effectively improve AUV navigation trajectories. However, its performance may depend on the quality of object detection. In environments characterized by strong acoustic noise or adverse underwater conditions, object detection may degrade or produce false positives, which can weaken the constraints in the pose graph and consequently

reduce the overall correction performance. To mitigate these effects, the proposed framework incorporates both artificial and natural objects as landmarks. Artificial objects generally exhibit more distinguishable acoustic characteristics than natural backgrounds, which may help reduce the likelihood of misdetection and limit the accumulation of localization errors. In addition, spurious detections are filtered through a distance-based temporal filtering scheme that leverages frame-to-frame consistency across consecutive sequences, which can suppress single-frame false positives that lack temporal continuity.

As shown in Fig. 11 (b), the sway-direction error exhibits a more pronounced improvement compared to the surge direction. Notably, sharp reductions in error are observed around 30 s and 100–200 s. Examining the corresponding sonar data revealed that a cylinder is present near 30 s, while a brick appears in the range of approximately 100–200 s. At these moments, the detection confidence for artificial objects was markedly higher compared to other time periods, leading to stronger pose-graph constraints and more significant error correction. This reveals that the presence of geometrically distinct objects—such as artificial structures, artificial reefs, or underwater volcanic features—intermittently within the survey area can dramatically improve localization accuracy compared to relying solely on natural objects as reference landmarks. Furthermore, the stable suppression of sway error observed in subsequent frames can be attributed to the continuous detection of seaweed across consecutive frames, where the distance-based temporal filtering effectively eliminated spurious detections and retained only reliable landmark observations.

The choice of detection-related parameters introduces inherent trade-offs in system performance. A lower confidence threshold increases the risk of incorporating false-positive constraints, whereas a higher threshold reduces available landmark observations. Similarly, an excessively large weight assigned to detection-based constraints amplifies sensitivity to detection errors, while an insufficiently small weight diminishes the contribution of sonar observations relative to odometry. To characterize these trade-offs, the relative ratio between odometry and detection constraint weights was varied systematically, and localization performance remained consistent within a reasonable neighborhood of the chosen parameters.

Regarding computational feasibility, the dominant processing cost of the proposed pipeline arises from the object detection stage. The YOLO-based detector achieved an average inference time of 38.65 ms per sonar frame (approximately 25.87 fps) when evaluated on an NVIDIA GeForce RTX 3060 desktop environment. Given the event-driven pose-graph optimization scheme, the overall processing demand remains bounded, indicating strong potential for real-time deployment on AUV platforms.

6 Conclusion

In this study, we presented a sim-to-real, detector-based pose-graph optimization framework for AUV localization using synthetic FLS imagery. Collecting and manually annotating large-scale real FLS datasets is time-consuming, costly, and operationally challenging. Most detector-based prior studies rely on artificial targets deployed in controlled test sites, thereby improving navigation only when man-made structures are visible in the sonar field of view. To overcome these limitations, we constructed a fully synthetic FLS dataset using FURo-sim, automatically generated bounding-box annotations, and defined a semantic label set. A YOLO-based sonar object detector was trained exclusively on these simulator-generated images and evaluated only on real FLS data, with the detected object centroids incorporated as landmark constraints in a 2-D pose graph that fuses DVL/IMU dead reckoning with detection-derived relative motion measurements. In the validation test, the proposed framework reduced the RMSE with respect to USBL from 0.716 m to 0.604 m corresponding to a relative reduction of approximately 15.4% compared with dead reckoning alone. These results indicate that a detector trained purely on synthetic sonar imagery can still improve real-world AUV navigation accuracy. Furthermore, under intentionally degraded dead-reckoning conditions with additional cumulative sway velocity noise, the framework consistently reduced RMSE by 21.2–25.4%, confirming its robustness against sensor drift even when odometry quality deteriorates.

Author Contributions J. Ha coordinated the study, designed the pose correction framework, implemented the synthetic data generation and training pipeline, and performed the main experiments and data analysis. H. Sim contributed to the development of the trajectory-correction algorithm and pose-graph optimization, and assisted with experiments and data analysis. H. Kweon developed the original FURo-sim virtual environment and the synthetic sonar dataset used in this work, and contributed to the design and refinement of the trajectory-correction scheme. H. Joe supervised the project, advised on the methodology and experimental design, and contributed to the interpretation of the results and revision of the manuscript. S.-c. Yu provided the coastal field datasets used in this study.

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Data Availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethics Approval Not applicable.

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References

1. Lee D, Kim D, Lee S, Myung H, Choi HT (2013) Experiments on localization of an AUV using graph-based SLAM. In: 2013 10th International conference on ubiquitous robots and ambient intelligence (URAI)
2. Jung J, Lee Y, Kim D, Lee D, Myung H, Choi HT (2017) AUV SLAM using forward/downward looking cameras and artificial landmarks. In 2017 IEEE Underwater technology (UT) IEEE, pp 1–3
3. Fallon MF, Kaess M, Johannsson H, Leonard JJ (2011) Efficient AUV navigation fusing acoustic ranging and side-scan sonar. In: 2011 IEEE International conference on robotics and automation IEEE, pp 2398–2405
4. Wang J, Shan T, Chandrasekaran M, Osedach T, Englot B (2019) Deep learning for detection and tracking of underwater pipelines using multibeam imaging sonar. In Proceedings of the IEEE international conference on robotics and automation workshop
5. Olivastri E, Fusaro D, Li W, Mosco S, Pretto A (2024) A sonar-based AUV positioning system for underwater environments with low infrastructure density. [arXiv:2405.01971](https://arxiv.org/abs/2405.01971)
6. Aubard M, Madureira A, Teixeira L, Pinto J (2025) Sonar-based deep learning in underwater robotics: overview, robustness, and challenges. *IEEE Journal of Oceanic Engineering*
7. Gomes LS, Almeida GP, Moreira BU, Quiroz M, Xavier B, Soares L, Brião SL, Oliveira FG, Drews-Jr PL (2025) Sonar image datasets: a comprehensive survey of resources, challenges, and applications. [arXiv:2510.03353](https://arxiv.org/abs/2510.03353)
8. Sung M, Kim J, Lee M, Kim B, Kim J, Yu SC (2020) Realistic sonar image simulation using deep learning for underwater object detection. *Int J Control Autom Syst* 18(3):523–534
9. Kamal Basha S, Nambiar A (2024) S3Simulator: a benchmarking side scan sonar simulator dataset for underwater image analysis. In International conference on pattern recognition, Springer, pp 219–235
10. Sung M, Cho H, Kim J, Yu SC (2019) Sonar image translation using generative adversarial network for underwater object recognition. In 2019 IEEE underwater technology (UT) IEEE, pp 1–6

11. Waldum AG, Pizarro O, Ludvigsen M (2023) Sonar object detection from synthetic sonar data generated by a sonar simulator and an adversarial neural network. In OCEANS 2023-Limerick IEEE, pp 1–10
12. Mascaro R, Teixeira L, Hinzmann T, Siegwart R, Chli M (2018) Gomsf: Graph-optimization based multi-sensor fusion for robust uav pose estimation. In 2018 IEEE international conference on robotics and automation (ICRA) IEEE, pp 1421–1428
13. Forsgren B, Brink K, Ganesh P, McLain TW (2023) Incremental cycle bases for cycle-based pose graph optimization. *IEEE Robot Automat Lett* 8(2):1021–1028
14. Grisetti G, Kümmerle R, Stachniss C, Burgard W (2010) A tutorial on graph-based slam. *IEEE Intell Transp Syst Mag* 2(4):31–43
15. Dong Z, Yan W, Cui R, Lei L, He Y (2024) An adaptive grouping sonar-inertial odometry for underwater navigation. *Ocean Eng* 294:116688
16. Xu Y, Zheng R, Zhang S, Liu M (2022) Robust inertial-aided underwater localization based on imaging sonar keyframes. *IEEE Trans Instrum Meas* 71:1–12
17. Kweon H, Sim H, Joe H (2025) Development of maritime robotics simulator using unreal engine 5. In: 2025 22nd International conference on ubiquitous robots (UR) IEEE, pp 597–602
18. Redmon J, Divvala S, Girshick R, Farhadi A (2016) You only look once: unified, real-time object detection. In Proceedings of the IEEE conference on computer vision and pattern recognition, pp 779–788
19. Paull L, Saeedi S, Seto M, Li H (2013) Auv navigation and localization: a review. *IEEE J Oceanic Eng* 39(1):131–149
20. González-García J, Gómez-Espinosa A, Cuan-Urquiza E, García-Valdovinos LG, Salgado-Jiménez T, Escobedo Cabello JA (2020) Autonomous underwater vehicles: localization, navigation, and communication for collaborative missions. *Appl Sci* 10(4):1256
21. Cao X, Ren L, Sun C (2022) Research on obstacle detection and avoidance of autonomous underwater vehicle based on forward-looking sonar. *IEEE Trans Neural Netw Learn Syst* 34(11):9198–9208
22. Islam MS (2025) Deep learning-based sonar image object detection system. *Int J Inf Syst Comput Eng (INJIISCOM)* 6(2):186–199
23. Sethuraman AV, Sheppard A, Bagoren O, Pinnow C, Anderson J, Havens TC, Skinner KA (2025) Machine learning for shipwreck segmentation from side scan sonar imagery: Dataset and benchmark. *Int J Robot Res* 44(3):341–354
24. Baatar G, Ritzau L, Karimanzira D, Rauschenbach T (2024) Enhanced underwater object detection in sonar images with range-setting embedding and sliding window preprocessing. In: OCEANS 2024-Halifax IEEE, pp 1–7
25. Pathak S, Uygur I, Shize L, Miyagusuku R, Moro A, Yamashita A, Asama H (2021) Localization in a semantic map via bounding box information and feature points. In 2021 IEEE/SICE International symposium on system integration (SII) IEEE, pp 126–131
26. Shin J, Chang S, Bays MJ, Weaver J, Wettergren TA, Ferrari S (2022) Synthetic sonar image simulation with various seabed conditions for automatic target recognition. In OCEANS 2022, Hampton Roads IEEE, pp 1–8
27. Sim H, Joe H (2024) Voronoi diagram-based USBL outlier rejection for AUV localization. *J Ocean Eng Technol* 38(3):115–123

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